Kwabena — here is your **Day 64 Dynamic Routing (RIP) Documentation**, fully structured, clean, and matching the style of Day 60–63. This is ready to paste directly into your repo as `day64-dynamic-routing.md`.

⭐ DAY 64 — Dynamic Routing (RIP) Documentation

1. Overview

Day 64 introduced **dynamic routing**, where routers automatically exchange network information instead of relying on manually configured static routes. I learned:

- How RIP (Routing Information Protocol) works
- How routers advertise networks to each other
- How routing tables update automatically
- How dynamic routing replaces static routing in multi-router networks
- How RIP Version 2 supports subnet masks and modern networks

This lab builds directly on Day 63's topology.

2. Network Topology

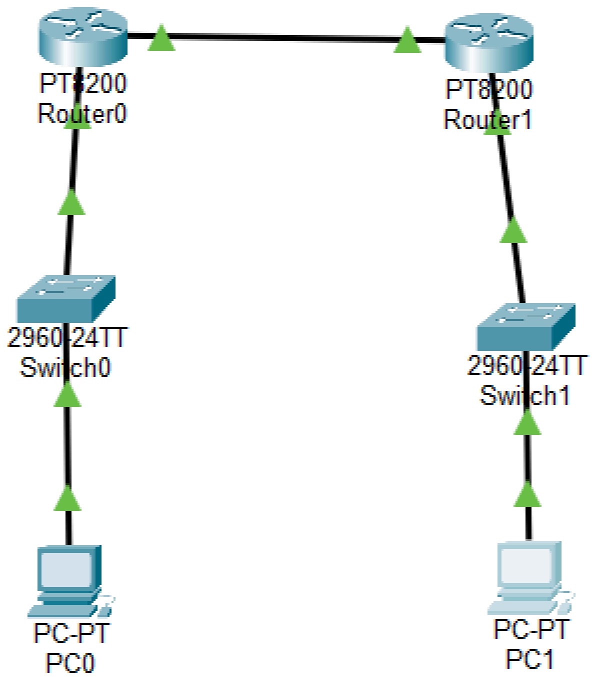
The topology is identical to Day 63:

Code
```
 Subnet A (LAN 1)            Subnet B (LAN 2)
[PC-A]                     [PC-B]
  |                          |
[SW-A]                       [SW-B]
  |                          |
[Router 1] ----------- [Router 2]
      Link Network: 10.0.0.0/30
```

📸 INSERT SCREENSHOT #1 — Day 64 Topology Here

Highlight:

- Router 1
- Router 2



-
- Switches
- PCs
- Router-to-router link

3. IP Addressing Plan

Same IP plan as Day 63:

LAN Subnets

Subnet	Network	Host Range	Gateway
Subnet A	192.168.10.0/26	1–62	192.168.10.1

Subnet B	192.168.20.0/26	1–62	192.168.20.1

Router-to-Router Link

Device	IP	Mask
Router 1 (G0/0/1)	10.0.0.1	255.255.255.252
Router 2 (G0/0/1)	10.0.0.2	255.255.255.252

PC IPs

PC	IP	Mask	Gateway
PC-A	192.168.10.10	255.255.255.192	192.168.10.1
PC-B	192.168.20.10	255.255.255.192	192.168.20.1

4. Removing Static Routes

Dynamic routing should not run alongside static routing. Before enabling RIP, static routes were removed:

Router 1:

Code

```
no ip route 192.168.20.0 255.255.255.192 10.0.0.2
```

Router 2:

Code

```
no ip route 192.168.10.0 255.255.255.192 10.0.0.1
```

📸 INSERT SCREENSHOT #2 — Static Route Removal Here

5. Enabling RIP (Dynamic Routing)

Router 1:

Code
router rip
version 2
network 192.168.10.0
network 10.0.0.0
no auto-summary

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip route 192.168.20.0 255.255.255.192 10.0.0.2
Router(config)#end
Router#
*Aug 31, 17:15:28.1515: SYS-5-CONFIG_I: Configured from console by console
Router#enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.10.0
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
*Aug 31, 17:15:52.1515: SYS-5-CONFIG_I: Configured from console by console
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
R    192.168.20.0/24 [120/1] via 10.0.0.2, 00:00:15, GigabitEthernet0/0/1

Router#
Router#
```

📸 INSERT SCREENSHOT #3 — Router 1 RIP Config Here

Router 2:

Code
router rip
version 2
network 192.168.20.0
network 10.0.0.0
no auto-summary

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip route 192.168.10.0 255.255.255.192 10.0.0.1
Router(config)#end
Router#
*Aug 31, 17:15:40.1515: SYS-5-CONFIG_I: Configured from console by console
Router#enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 192.168.20.0
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
*Aug 31, 17:16:03.1616: SYS-5-CONFIG_I: Configured from console by console
```

📸 INSERT SCREENSHOT #4 — Router 2 RIP Config Here

6. Verifying RIP Routing Tables

After enabling RIP, both routers automatically learned each other’s LAN networks.

Router 1 — `show ip route`

Expected dynamic route:

Code
R 192.168.20.0/26 [120/1] via 10.0.0.2

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
R    192.168.20.0/24 [120/1] via 10.0.0.2, 00:00:15, GigabitEthernet0/0/1

Router#
Router#
```

📸 INSERT SCREENSHOT #5 — Router 1 Routing Table Here

Router 2 — `show ip route`

Expected dynamic route:

Code
R 192.168.10.0/26 [120/1] via 10.0.0.1

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0/1
     192.168.10.0/26 is subnetted, 1 subnets
R       192.168.10.0 [120/1] via 10.0.0.1, 00:00:01, GigabitEthernet0/0/1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/26 is directly connected, GigabitEthernet0/0/0
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0

Router#
Router#
```

📸 INSERT SCREENSHOT #6 — Router 2 Routing Table Here

7. End-to-End Ping Tests

Dynamic routing should allow full communication between both LANs.

From PC-A:

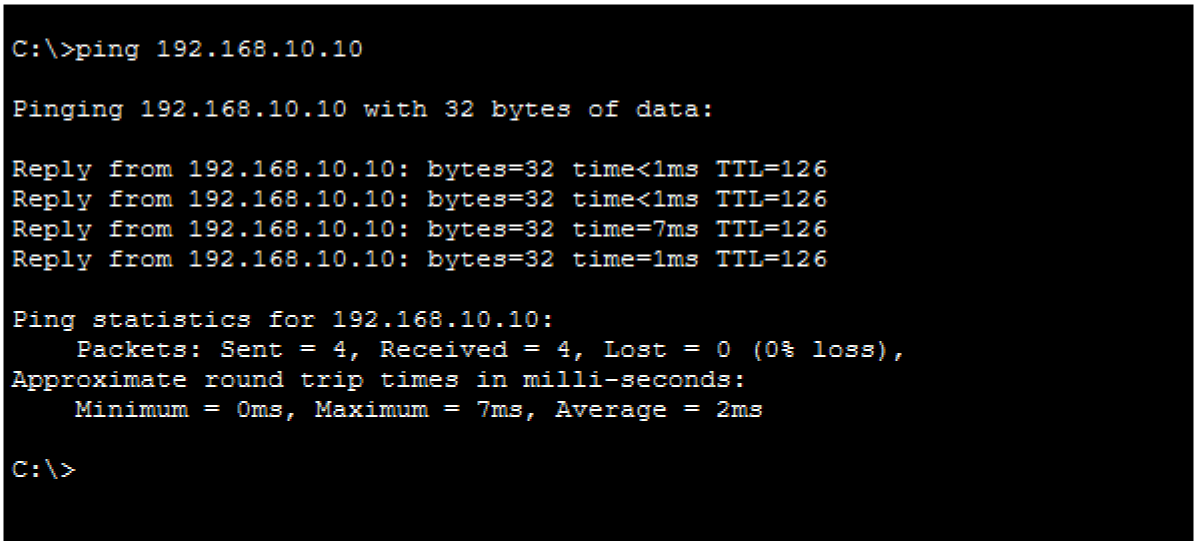
Code
ping 192.168.20.10
Reply from 192.168.20.10: bytes=32 time<1ms TTL=128

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

📸 INSERT SCREENSHOT #7 — PC-A Ping Test Here

From PC-B:

Code
ping 192.168.10.10
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128



```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time=7ms TTL=126
Reply from 192.168.10.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>
```

📸 INSERT SCREENSHOT #8 — PC-B Ping Test Here

8. How RIP Works (Explanation)

RIP is a distance-vector routing protocol. It works by:

1. **Advertising networks** to neighboring routers
2. **Receiving network advertisements** from neighbors
3. **Updating routing tables** based on hop count
4. **Choosing the shortest path** (fewest hops)
5. **Automatically adapting** when links change

RIP Version 2 supports:

- subnet masks
- classless routing
- modern network designs

This makes RIP suitable for small to medium networks.

9. What I Learned

- How dynamic routing simplifies multi-router networks
- How RIP automatically discovers remote networks
- How routing tables update without manual configuration
- How RIP Version 2 supports subnet masks
- How dynamic routing scales better than static routing
- How enterprise networks avoid manual route management

Dynamic routing made the network feel alive — routers communicate and learn automatically.

10. Reflection

Day 64 felt like a major step forward. Static routing taught me how routers think manually, but RIP showed me how routers think **on their own**. Seeing routes appear automatically in the routing table made dynamic routing feel powerful and intuitive.

This lab made me feel like I'm building real enterprise networks.

⭐ Your Next Step

Choose what you want next:

- **Start Day 65 — RIP with 3 Routers**
- **Explain RIP visually**
- **Do dynamic routing practice problems**